

SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE NAME: MLA03

COURSE CODE: Reinforcement learning for Environment and Agent

COURSE OUTCOME COVERED

***CO2:** Apply Dynamic Programming methods to solve reinforcement learning problems and derive optimal policies for decision-making tasks. (BL2, BL3)*

Assessment Tool Weightage: 30%

Dr G.A. SENTHIL

QUESTIONS: 10

Total Marks: 50

Annexure A
Application - Based Questions

Duration: 2Hrs

1. Design and develop a Reinforcement Learning framework a ride-sharing company uses Reinforcement Learning to match drivers with passengers efficiently. Apply RL concepts to define the state space, action space, and reward function to minimize waiting time and improve service quality. (5 Marks)

Objective

To efficiently match drivers with passengers, minimize passenger waiting time, reduce travel distance, and improve customer satisfaction.

Why Reinforcement Learning?

Reinforcement Learning is suitable because traffic conditions, driver availability, and passenger requests continuously change. The RL agent learns the best driver-passenger matching strategy from experience and improves over time.

Environment

Ride-sharing platform.

Agent

Ride-sharing matching system.

State Space

- Driver location
- Passenger request
- Traffic conditions
- Driver availability

Action Space

- Assign a driver
- Reject request
- Reassign nearby driver

Reward Function

- +20 for quick pickup
- +15 for customer satisfaction
- -10 for long waiting time
- -5 for unnecessary travel

Policy

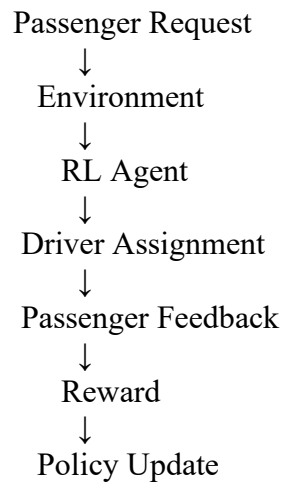
Select the available driver that maximizes reward.

Learning Process

The RL agent observes the current state, assigns a driver, receives a reward based on service quality, updates its policy, and repeats the process. Over time, it learns better matching strategies.

- The agent stores previous ride experiences.
- It updates its policy after every completed ride.
- The policy gradually learns which driver assignment produces the highest customer satisfaction.

RL Architecture



Expected Outcomes

- Reduced passenger waiting time
- Efficient driver utilization
- Improved service quality
- Higher customer satisfaction

2. An e-learning platform uses Reinforcement Learning to recommend personalized study materials. Recall how exploration and exploitation strategies help balance familiar and new content recommendations. (5 Marks)

Objective

To recommend personalized study materials that improve student learning performance.

Why Reinforcement Learning?

Students have different learning patterns. RL continuously learns from student interactions and recommends the most suitable study material.

- Every student learns differently.
- RL personalizes recommendations based on individual learning progress.
- It continuously updates recommendations using quiz and assignment performance.

State Space

- Student performance
- Completed lessons
- Quiz scores
- Learning history

Action Space

- Recommend video
- Recommend quiz
- Recommend notes

Reward Function

- +10 for course completion
- +8 for high quiz scores
- -5 for skipped lessons

Policy

Recommend the material with the highest expected learning improvement.

Exploration vs Exploitation

Exploration

- Recommend new learning materials
- Introduce new topics

Exploitation

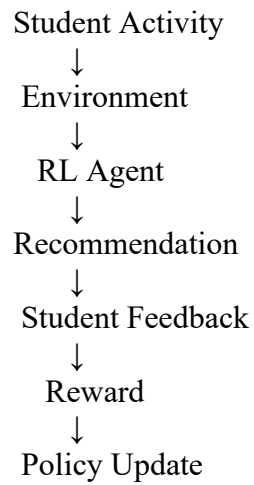
- Recommend materials similar to previous successful recommendations
- Focus on weak subjects

Learning Process

The agent receives student feedback, updates recommendations, and gradually improves learning outcomes.

- Student feedback helps update future recommendations.
- The system gradually identifies each student's strengths and weaknesses.
- Recommendations become increasingly personalized over time.

RL Architecture



Expected Outcomes

- Personalized learning
- Better exam scores
- Higher student engagement
- Improved knowledge retention
- Higher course completion rates
- Increased student motivation
- Personalized learning experience

3. Design and develop a Reinforcement Learning framework a robotic vacuum cleaner uses Reinforcement Learning to clean efficiently while avoiding obstacles. Apply the concept of policy and explain how the value function improves performance over time. (5 Marks)

Objective

Clean the maximum area while avoiding obstacles and minimizing energy consumption.

Why Reinforcement Learning?

The robot learns efficient cleaning paths from previous experiences without requiring predefined rules.

- ☐ The environment changes frequently due to moving furniture and obstacles.
- ☐ RL helps the robot adapt without manual programming.
- ☐ It continuously improves cleaning strategies through experience.

Environment

House with rooms and obstacles.

Agent

Robot vacuum cleaner.

State Space

- Robot position
- Dirt level
- Battery level
- Obstacle locations

Action Space

- Move forward
- Turn left
- Turn right
- Clean

Reward Function

- +10 for cleaning
- -10 for collision
- -2 for unnecessary movement

Policy

Choose actions that maximize cleaning efficiency.

Value Function

The value function estimates the future reward of each state.

Higher value → Better cleaning path.

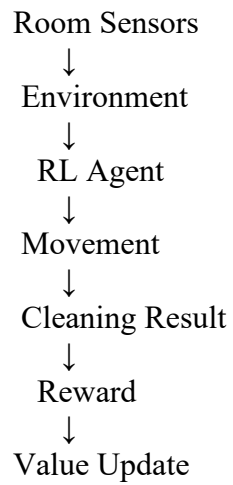
Lower value → Inefficient path.

Learning Process

The robot learns better routes after every cleaning session.

- The robot remembers successful cleaning paths.
- Value functions estimate which locations provide better future rewards.
- Cleaning efficiency improves after each cleaning cycle.

RL Architecture



Expected Outcomes

- Faster cleaning
- Reduced battery usage
- Better obstacle avoidance
- Complete room coverage
- Longer battery life
- Reduced cleaning time
- Better navigation efficiency

4. A cryptocurrency trading system uses Reinforcement Learning to decide buy, sell, or hold actions. Outline the challenges in designing reward functions and explain how poor reward design can affect performance. (5 Marks)

Objective

Maximize trading profit while minimizing financial risk.

Why Reinforcement Learning?

The cryptocurrency market changes continuously. RL adapts trading decisions based on market conditions.

- ☐ Cryptocurrency prices change rapidly.
- ☐ RL continuously updates trading strategies.
- ☐ It learns profitable trading patterns from market data.

State Space

- Current price
- Previous price
- Trading volume
- Market trend

Action Space

- Buy
- Sell
- Hold

Reward Function

- +20 for profitable trades
- -20 for losses
- +5 for successful holding decisions

Impact of Poor Reward Design

- Excessive risk-taking
- Frequent unnecessary trades
- Poor investment performance

Reward Design Challenges

- Immediate profit may not indicate long-term success.
- Poor reward design may encourage risky trading.
- Rewards should balance profit and risk.

Learning Process

The agent continuously updates its trading policy based on previous trading outcomes.

- The agent updates its strategy after every trade.
- It gradually identifies profitable market patterns.
- Long-term trading performance improves with experience.

RL Architecture



Expected Outcomes

- Better trading decisions
- Increased long-term profit
- Reduced financial risk
- Stable investment strategy
- Reduced financial losses
- Better portfolio management
- Increased profitability

5. Develop a smart grid system uses Reinforcement Learning to manage electricity distribution. Create an RL model by defining state space, action space, and reward mechanism.

(5 Marks)

Objective

Efficiently distribute electricity, reduce power wastage, and maintain a stable power supply.

Why Reinforcement Learning?

Electricity demand changes throughout the day. RL continuously learns from demand patterns and optimizes power distribution in real time.

- ☐ Electricity demand changes continuously.
- ☐ Renewable energy production is uncertain.
- ☐ RL adapts distribution decisions using real-time information.

State Space

- Electricity demand
- Battery level
- Renewable energy availability

Action Space

- Increase supply
- Reduce supply
- Store energy

Reward Function

- +20 for balanced distribution
- -15 for overload
- -10 for energy waste

Policy

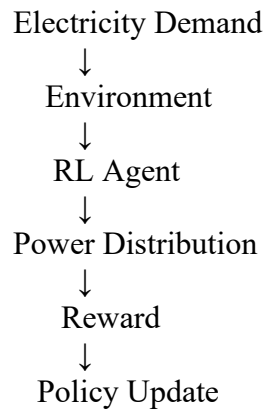
Select actions that maximize efficiency and reliability.

Learning Process

The RL agent observes electricity demand, selects the best action, receives a reward, updates its policy, and continuously improves distribution decisions.

- The RL agent learns demand patterns throughout the day.
- Policy updates improve future electricity allocation.
- Continuous learning enables better handling of peak demand periods.

RL Architecture



Expected Outcomes

- Reduced power wastage
- Better electricity distribution
- Improved grid stability
- Lower operational cost
- Increased renewable energy usage
- Lower electricity costs
- Reduced carbon emissions
- Higher system reliability

Code of Conduct:

I _S.Sirivally / 192472371_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Sirivally.S

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Problem Context	20	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand